Lab Report 4: Single-Photon Anti-Bunching

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Experimental Setup Used: #2

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1 Abstract

In the experiment, our group analyzed the quantum nature of light by measuring the $g^{(2)}(0)$ heralded photon emission using spontaneous down conversion (SPDC) to generate pairs. Our setup consisted of three single-photon detectors (SPADs) and a 50:50 beam splitter. The SPADs detected two-fold and three-fold coincidences at different power levels and time intervals. After accounting for background and noise, we found that $g^{(2)}(0) < 0.5$, with a minimum of approximately 0.13, showing strong antibunching behavior. The results from this lab support the interpretation of light being quantized particles.

2 Introduction

The quantum nature of light is one of the most critical discoveries in physics. During the early 1900's scientists discovered the photoelectric effect, which showed that light arrives in packets, and photon correlation experiments which showed light behaves as indivisible particles.

One key experimental method for analyzing the quantum nature of light is the measurement of the second-order function $g^{(2)}(\tau)$. This function measures the likelihood of detecting two photons separated by the time delay τ. Specifically, $g^{(2)}(0)$ shows us the probability of detecting two photons simultaneously. Classical light sources such as coherent or thermal sources, exhibit $g^{(2)}(0) \geq 1$, while true single photon sources exhibit $g^{(2)}(0) < 1$. An ideal single photon source would be $g^{(2)}(0) = 0$.

This lab aims to demonstrate this antibunching nature by evaluating the second-order function $g^{(2)}(0)$ of a heralded single-photon source. We will use the data from three different SPADs and measure coincidence rates along different power and interval lev
els using the following relation:

$$g^{(2)}(0) = \frac{P_{s1,s2}}{P_{s1}P_{s2}} = \frac{N_{s1,s2,i} \cdot N_i}{N_{s1,i} \cdot N_{s2,i}} \quad (1)$$

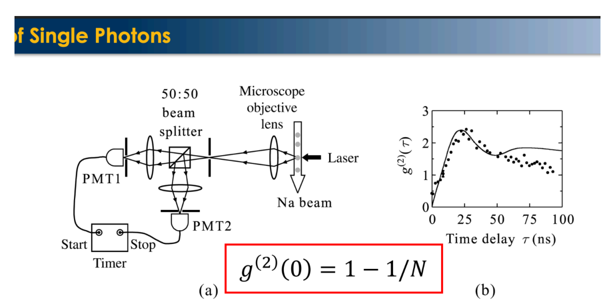

Figure 1: Experimental setup and results from first observation of photon bunching, showing $g^{(2)}(0) < 1$ from lecture slides, pg. 25

Providing a little more background, we will analyze the antibunching behavior by generating light through the SPDC process, a nonlinear optical process where a high-energy photon is split into a pair of down-converted photons called the signal and idler pair. By detecting the signal photon with a SPAD, we will "herald" the presence of a correlated idle photon. This heralded detection allows us to control measurements and detect external background noise/ uncorrelated events. We will then

examine if the idler photon is truly indivisible
(an individual photon).

3 Procedure

In this experiment, we will measure the second order correlation function $g^{(2)}(0)$ using a heralded single photon. The source will be created through SPDC. The setup consists of a 405 nm pump laser, a nonlinear optical crystal, optical fibers for coupled photons, a 50:50 beam splitter, and three SPAD detectors. The setup will allow us to measure one, two-fold, and threefold coincidences to calculate the correlation function.

First, the pump laser will enter a nonlinear crystal and produce two down converted photons in a pair. Through the optical fibers the signal photon will be “heralded” at the first SPAD detector. The idler photon will then be redirected through optical fibers to a 50:50 beam splitter, whose outputs are connected to two SPADs. This allows us to monitor if both detectors click for a herladed event.

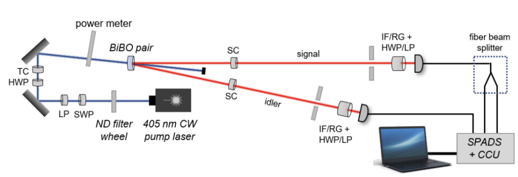

Figure 2: Experimental setup for heralded photon source using a 50:50 beam splitter and three detectors from tutorial4, pg. 4

When we are measuring the coincidence measurements there are three different values. Single coincidences are from all detectors. Two-fold coincidences are between: signal and SPAD1(idler 1) and signal and SPAD2(idler 2). Threefold coincidences are among the SPAD1, SPAD2, and signal. Measurements should be taken through three acquisition periods: 10s, 20s, and 30s to account for consistency. To study how power affects our measurements, we will use a ND filter wheel to reduce the beam intensity. In our experiment, the first measurements will be taken at power level 1 with varying backgrounds. Then, for the second part of the experiment we will measure the correlation function across 4 different beam intensities.

For the first part we will collect data at three different acquisition times (10s, 20s, 30s). These counts are then used to calculate the following:

$$g^{(2)}(0) = \frac{N_{s1,s2,i} \cdot N_i}{N_{s1,i} \cdot N_{s2,i}} \quad (2)$$

To account for background and noise for the first part of the experiment, we repeat the measurements for four different scenarios:

1. Lights on, laser on
2. Lights on, laser off
3. Lights off, laser blocked
4. Lights off, laser on

This allows for a more precise characterization of antibunching within quantum light, allowing for clear separation between background noise and true measurements.

False threefold coincidences are measured statistically using the following formula:

$$N_{acc} = (N_{s1,s} * N_{s2} + N_{s2,s} * N_{s1}) \cdot \Delta t \quad (3)$$

The true number of coincidences can be found by subtracting accidental and noise from the initial measurements and finding the proper correlation factor. Additionally, poissonian statistics will be assumed, allowing for error correction and find the uncertainties using this formula:

$$\Delta g^{(2)}(0) = g^{(2)}(0) \cdot \sqrt{\frac{1}{N_{s1,s2,s}} + \frac{1}{N_s} + \frac{1}{N_{s1,s}} + \frac{1}{N_{s2,s}}} \quad (4)$$

For the second part of the experiment, pump power is varied using the ND filter wheel through settings 1-4. Steps will then be repeated, except this time only measuring the values for the specific time interval of 30 seconds and measuring the same one fold, two-fold, and threefold coincidences. The same analysis will be repeated on the data and as well as the uncertainties.

4 Results and Discussion

4.1 Single-Photon Antibunching

First, we recorded the counts on each detector and their coincidence counts for each of the four scenarios listed in section 3. Values for the second-order correlation function at zero delay, $g^{(2)}(0)$, as well as its uncertainty, were then calculated for the cases in which the laser was on.

	10s	20s	30s
N_s	29656	29384	29395
N_{i1}	63828	66362	66458
N_{i2}	25840	26155	26459
$N_{s,i1}$	246	232	238
$N_{s,i2}$	151	143	140
$N_{s,i1,i2}$	0.9	0.9	1

Table 1: Counts-per-second versus acquisition time taken with the laser unblocked and lights on.

Using equations (2) and (4), we obtained the following second-order correlation functions for each acquisition time:

10s: $g^{(2)}(0) = 0.719 \pm 0.241$

20s: $g^{(2)}(0) = 0.797 \pm 0.189$

30s: $g^{(2)}(0) = 0.882 \pm 0.162$

This was then repeated for the other three scenarios, with the correlation function calculated again for the case in which the laser was on. In the case in which the three-fold coincidence count was measured as zero, the counts-per-second was substituted as one over the acquisition time.

	10s	20s	30s
N_s	22357	22165	21971
N_{i1}	4902	4912	4934
N_{i2}	3938	3892	3899
$N_{s,i1}$	148	144	144
$N_{s,i2}$	115	117	117
$N_{s,i1,i2}$	.1	.15	.1

Table 2: Counts-per-second versus acquisition time taken with the laser unblocked and lights off.

With the lights turned off, the second-order correlation functions decreased dramatically:

10s: $g^{(2)}(0) = 0.131 \pm 0.131$

20s: $g^{(2)}(0) = 0.197 \pm 0.114$

30s: $g^{(2)}(0) = 0.130 \pm 0.014$

	10s	20s	30s
N_s	9025	9225	9246
$N_{s,i1}$	28	23	26
$N_{s,i2}$	8	7	10
$N_{s,i1,i2}$	362	364	366

Table 3: Counts-per-second versus acquisition time taken with the laser blocked and lights on.

	10s	20s	30s
N_s	378	368	368
$N_{s,i1}$	0	0	0
$N_{s,i2}$	0	0	0
$N_{s,i1,i2}$	1/10	1/20	1/30

Table 4: Counts-per-second versus acquisition time taken with the laser blocked and lights off.

4.2 Photon Statistics

We notice that the three-fold coincidence count is non-zero, due to contamination from outside light sources, which can skew the data. *Eqn. 3* is used to verify the number of accidental coincidence counts where Δt is the time window which is when photons are detected; its given value is 25ns.

$$\text{10s: } N_{acc} = 0.39 \pm 0.02$$
$$\text{20s: } N_{acc} = 0.38 \pm 0.02$$
$$\text{30s: } N_{acc} = 0.157 \pm 0.009$$

Finding the two-fold accidental coincidence count can be given by:

$$N_{acc} = N_i \cdot N_s \cdot \Delta t$$

Where N_i is the cps for the given idler count, N_s is the cps for the idler photons, and Δt is the same value given for the fold accidental count.

N_{acc}	10s	20s	30s
s, i1	47.29	48.72	48.84
s, i2	19.14	19.2	19.42
s, i1, i2	0.40	0.39	0.39

Table 5: Accidental Coincidence Counts for 2-fold and 3-fold Coincidence Counts(lights on)

Through these values, the $g^{(2)}(0)$ and $\Delta g^{(2)}(0)$ can be corrected to take into consideration outside light sources.

$$\text{10s: } g^{(2)}(0) = 0.399 \pm 0.134$$
$$\text{20s: } g^{(2)}(0) = 0.453 \pm 0.152$$
$$\text{30s: } g^{(2)}(0) = 0.538 \pm 0.171$$

Similarly, corrected values for the case in which the lights were off were calculated:

N_{acc}	10s	20s	30s
s, i1	2.74	2.722	2.71
s, i2	2.201	2.157	2.142
s, i1, i2	0.029	0.0028	0.028

Table 6: Accidental Coincidence Counts for 2-fold and 3-fold Coincidence Counts(lights off)

$$\text{10s: } g^{(2)}(0) = 0.094 \pm 0.094$$
$$\text{20s: } g^{(2)}(0) = 0.160 \pm 0.131$$
$$\text{30s: } g^{(2)}(0) = 0.093 \pm 0.093$$

These values show the existence of antibunching, showcasing the fact that light is quantized.

4.3 Power Dependence

The experiment from Section 4.1 was then repeated, varying the pump power rather than the now constant 30-second acquisition time.

	2	3	4
N_s	18039	14898	10198
N_{i1}	4095	3366	2380
N_{i2}	3170	2723	1885
$N_{s,i1}$	116	89	64
$N_{s,i2}$	93	81	53
$N_{s,i1,i2}$	0.1	0.1	0.1

Table 7: Counts-per-second versus pump power setting

Using equations (2) and (4), the following second-order correlation functions were calculated:

Power Setting 2: $g^{(2)}(0) = 0.167 \pm 0.097$
Power Setting 3: $g^{(2)}(0) = 0.207 \pm 0.119$
Power Setting 4: $g^{(2)}(0) = 0.301 \pm 0.174$

Due to the Poissonian statistics of light, we would expect a higher chance of bunching as power is increased, due to Poissonian noise being signal-level dependent noise. This is consistent with our calculated values for $g^{(2)}(0)$ versus the pump power. More specifically, we see an increasing trend as the pump power is increased, as we expected.

5 Conclusion

In this lab, we measured the second-order correlation function $g^{(2)}(0)$ to investigate the quantum nature of our photon source. After correcting for background noise and accidental coincidences, our measurements showed $g^{(2)}(0) < 0.5$, confirming single-photon antibunching and the quantum nature of our light source. Lower pump power showed stronger antibunching, as expected.